WEALTH CORE

SYSTEMIC ZERO + PERSISTENT BEAR

Investment Prospectus & Engineering Specification

Provisional Implementation Champion | True Next-Open Validation

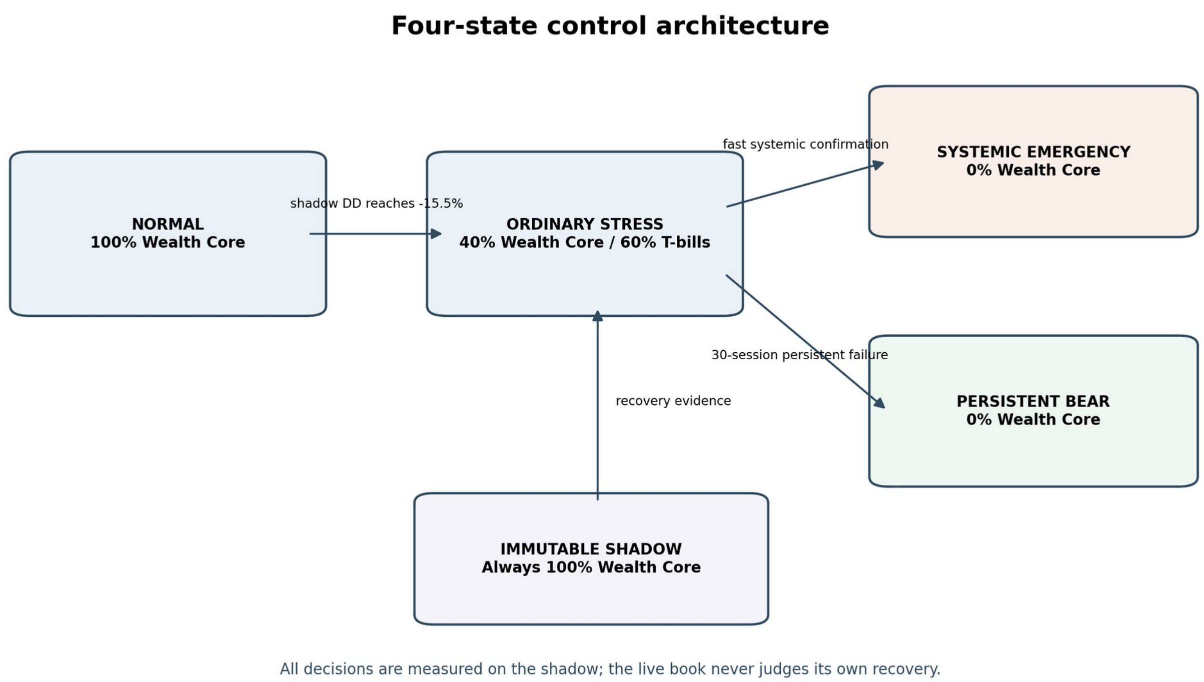

Figure 1. The champion combines continuous equity selection with a shadow-governed risk controller.

Research status

This document is a technical and investment research prospectus. It is not an offering memorandum, does not constitute investment advice, and does not represent audited live performance. Historical simulations can differ materially from live results.

Specification date: August 6, 2026

Prepared for engineering implementation, simulator certification, and investment review

Contents

How to read this document

Sections 1-5 explain the investment proposition. Sections 6-14 specify the system behavior. Sections 15-18 define risks, tests, configuration, and terminology.

1. Prospectus at a glance

Measure	Champion	SPY	QQQ	VTI
20-year CAGR	20.30%	11.26%	16.64%	11.20%
Maximum drawdown	-28.64%	-55.20%	-53.41%	-55.45%
Ending value of $1M	$40.33M	$8.44M	$21.72M	$8.36M
Annualized volatility	18.09%	19.43%	22.09%	19.60%

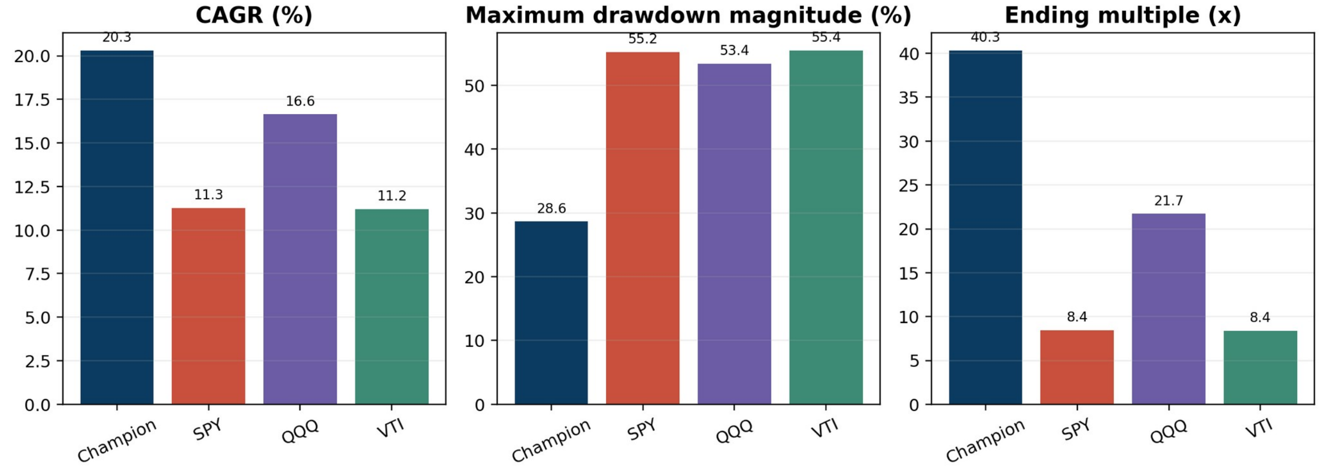

Figure 2. Headline metrics use the same July 31, 2006 through July 31, 2026 period.

Core proposition

Pursue long-horizon equity compounding while using an immutable shadow portfolio to identify ordinary stress, fast systemic shocks, and prolonged periods in which remaining equity exposure is no longer creating wealth.

What is new

- The systemic emergency floor is reduced from 40% equity to 0%.
- A Persistent Bear state moves from 40% to 0% only after at least 30 sessions of sustained failure.
- The positive-day metric was removed because it did not change any historical decision.
- All transitions were retested using actual next-session Sharadar opening prices.

2. Investment thesis

Wealth Core is designed to own durable winners long enough for exceptional compounding to matter. Its principal weakness is that a concentrated portfolio of past winners can experience severe synchronized drawdowns when leadership breaks. The defensive controller is therefore not a separate alpha engine. It is a capital-preservation overlay that changes exposure only when the strategy itself demonstrates broad impairment.

Three ideas

Idea	Implication
Let winners compound	Avoid routine turnover and prediction-driven exits.
Measure stress on a full-exposure shadow	Risk signals remain comparable through time and cannot be distorted by prior de-risking.
Use different logic for crashes and grinding bears	Fast shocks and prolonged negative expectancy require distinct entry tests but share a 0% defensive state.

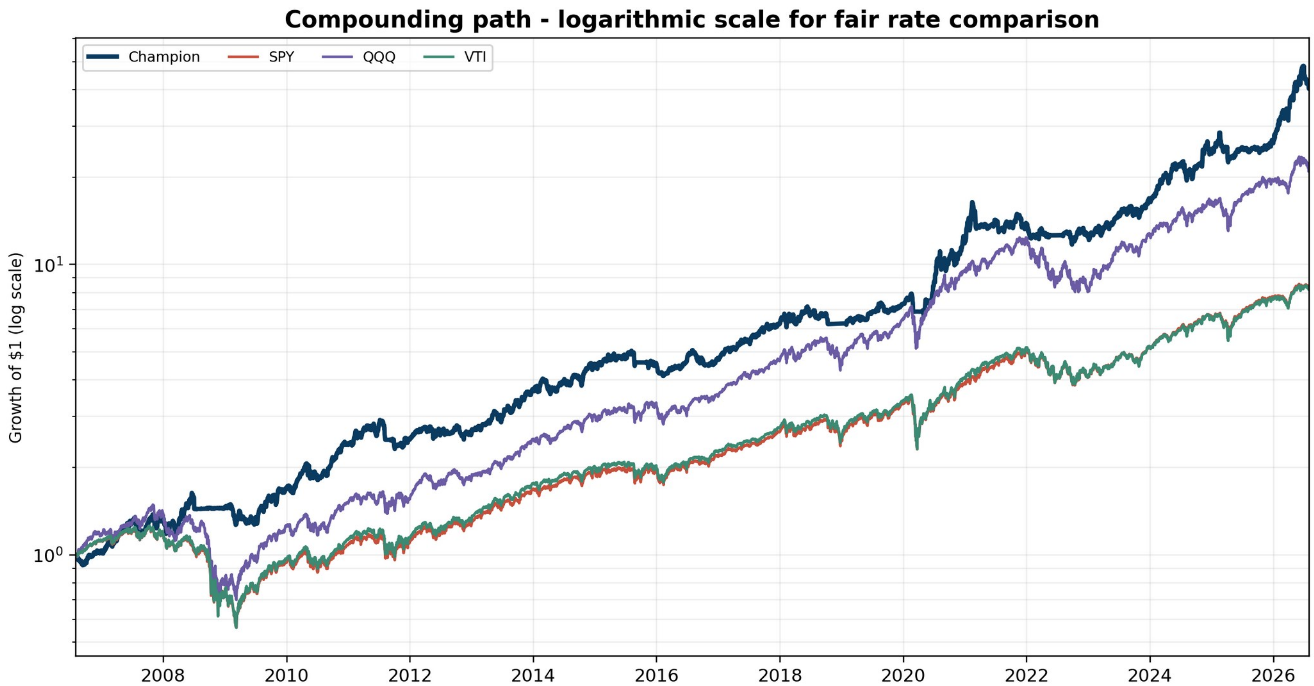

Figure 3. Log-scale compounding comparison; identical dates and one shared axis.

The log-scale chart is the fairest visual comparison of growth rates. Equal vertical distances represent equal percentage changes. The linear chart in the next section shows the economic magnitude of ending wealth.

3. Strategy architecture

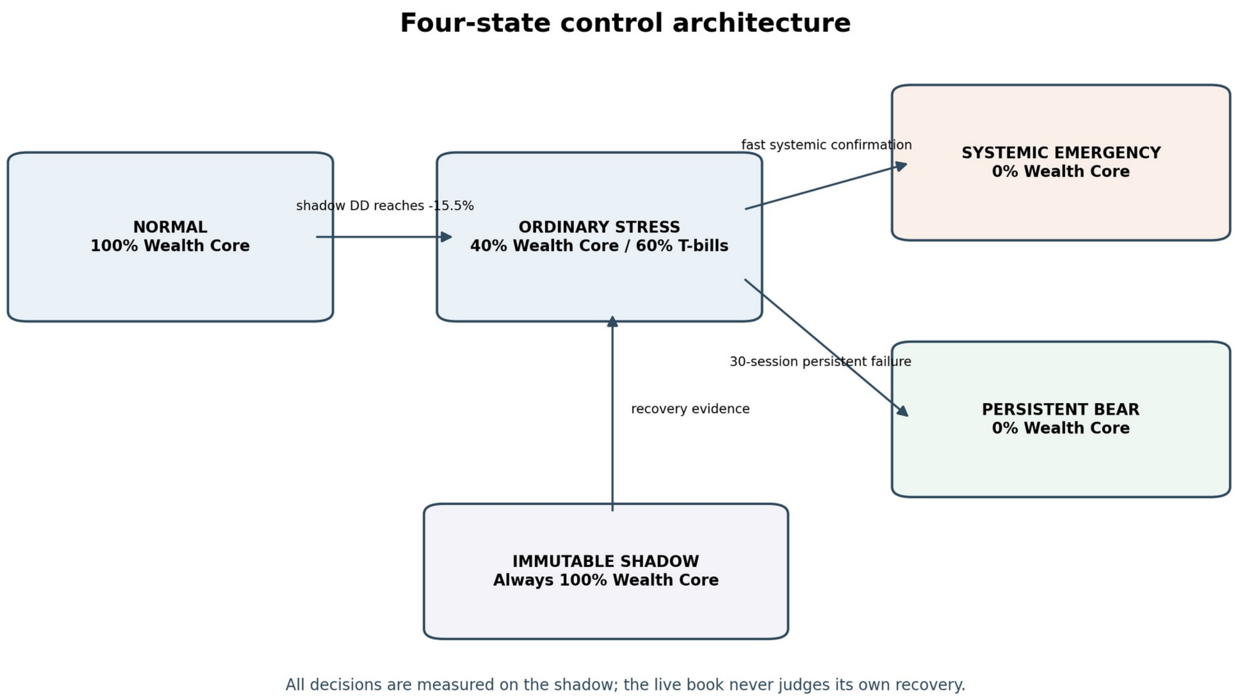

Figure 4. Four-state controller and immutable shadow control plane.

State	Wealth Core	T-bills	Purpose
NORMAL	100%	0%	Full participation
ORDINARY_STRESS	40%	60%	Reduce exposure after 15.5% shadow drawdown
SYSTEMIC_EMERGENCY	0%	100%	Exit during confirmed rapid systemic stress
PERSISTENT_BEAR	0%	100%	Exit after sustained negative shadow expectancy

Design invariants

- Wealth Core selection logic continues to run in every state.
- The shadow remains fully invested and receives all simulated fills and corporate actions.
- Live allocation never changes shadow holdings, entry ages, peaks, cooldowns, or pending orders.
- Recovery decisions are based on the shadow, not the protected live portfolio.
- Systemic emergency has priority over Persistent Bear when both qualify.

4. Performance and benchmark comparison

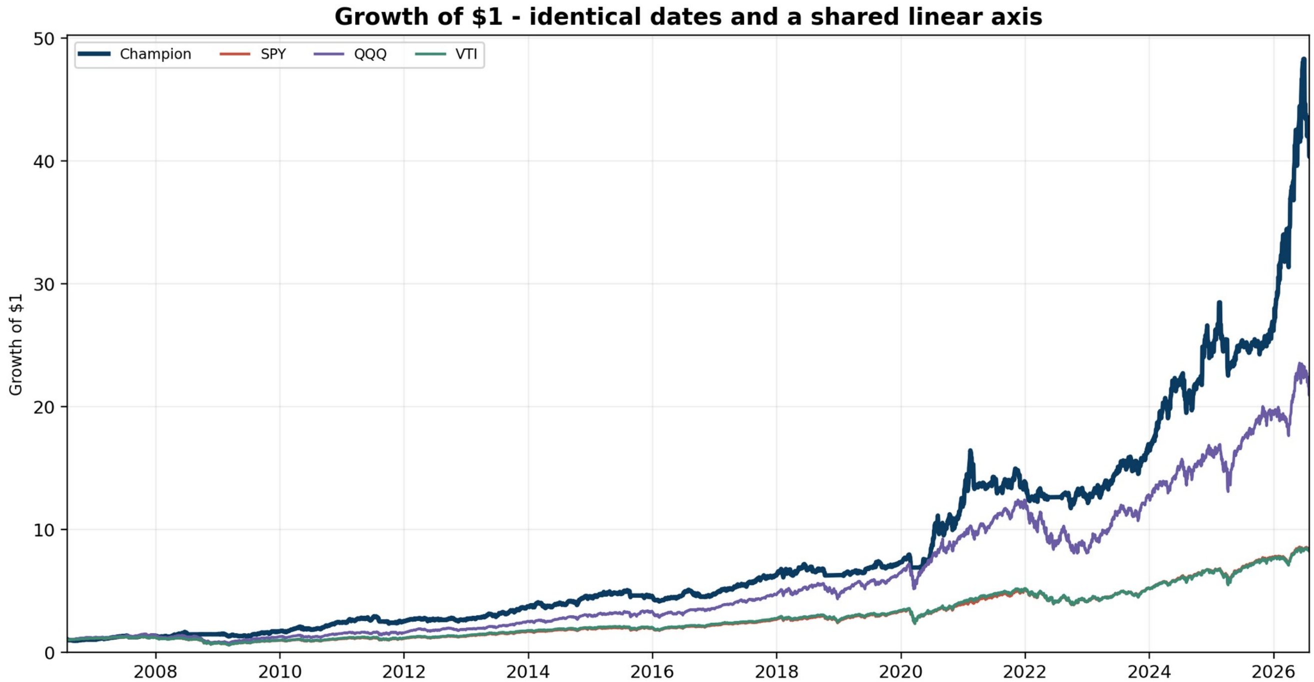

Figure 5. Growth of $1 on a shared linear axis. No series uses an independent scale.

The linear chart makes ending wealth visible but naturally compresses lower-growth benchmark paths. Figure 3 should be used when comparing growth rates; Figure 5 should be used when comparing economic outcomes.

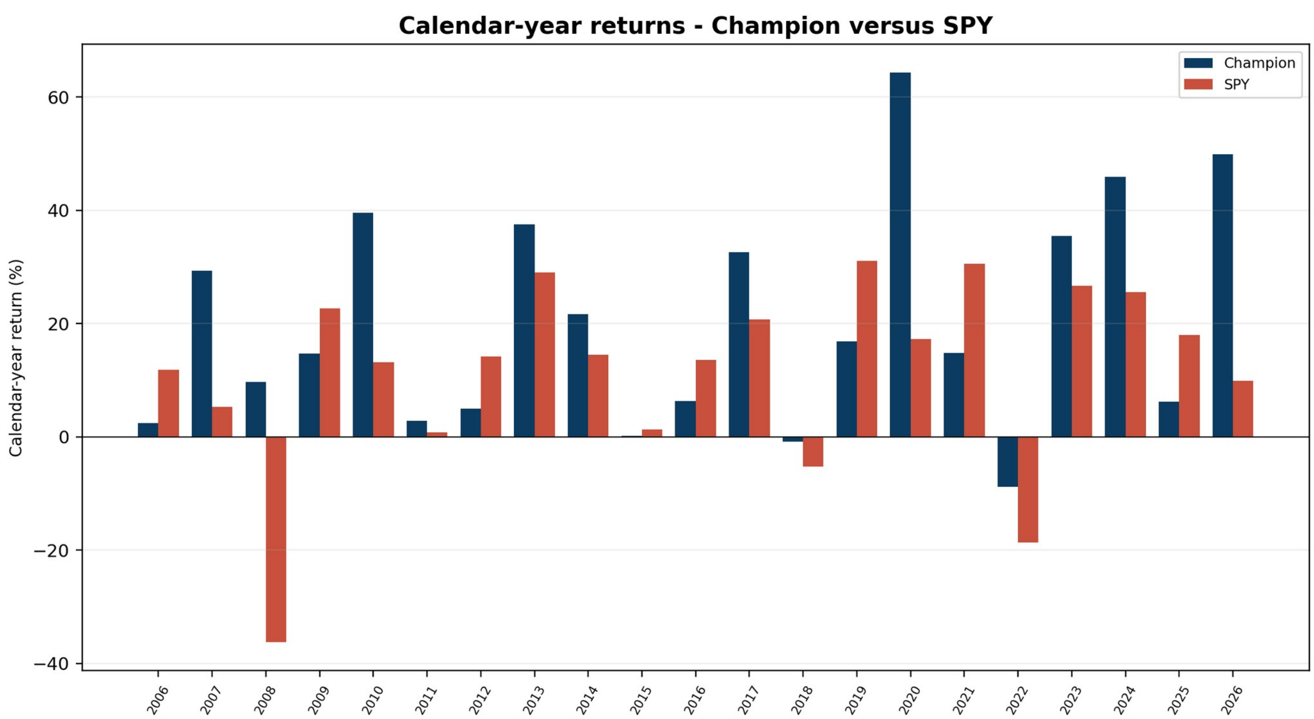

Figure 6. Calendar-year returns for the champion and SPY.

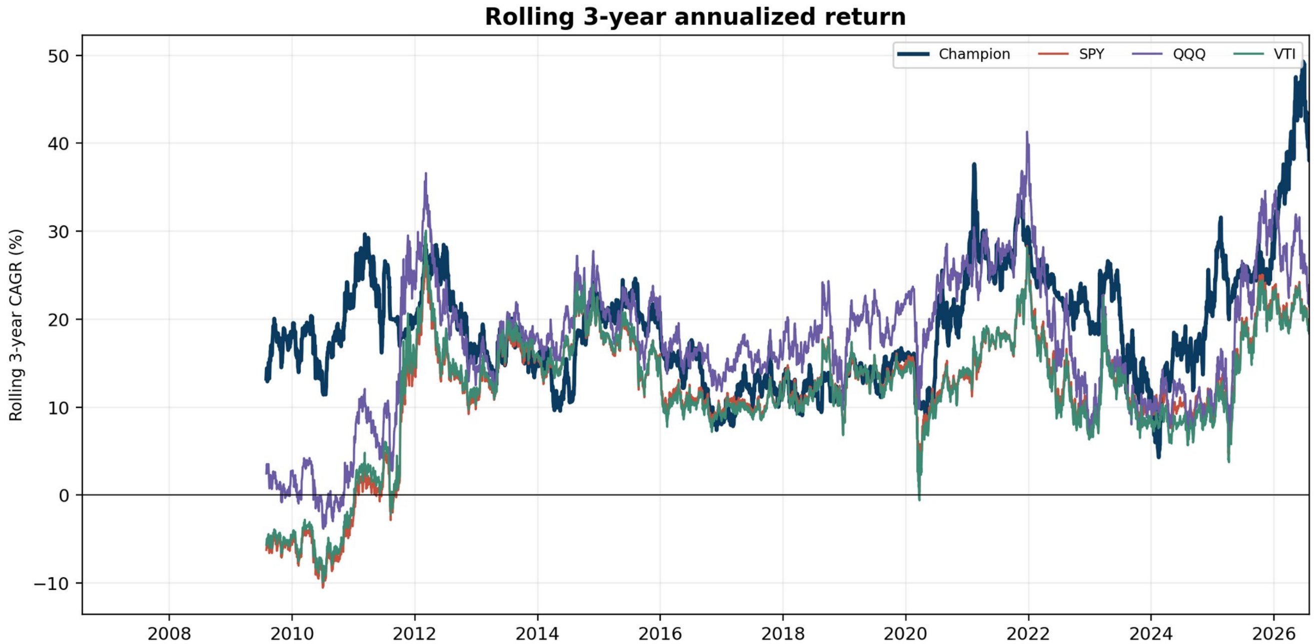

Figure 7. Rolling three-year CAGR, calculated on identical 756-session windows.

Benchmark interpretation

- SPY represents the capitalization-weighted US large-cap market.
- QQQ represents a highly regarded growth-oriented Nasdaq-100 fund and is a demanding upside benchmark.
- VTI represents the broad US investable equity market.
- All three benchmark series are total-return-adjusted fund-price series from the same Sharadar SFP dataset.
- No benchmark is rescaled independently within a comparison chart.

Result

Under the true next-open overlay, the champion produced a 20.30% CAGR, a -28.64% maximum drawdown, and a 40.33x ending multiple over the 20-year interval.

5. Risk behavior and stress periods

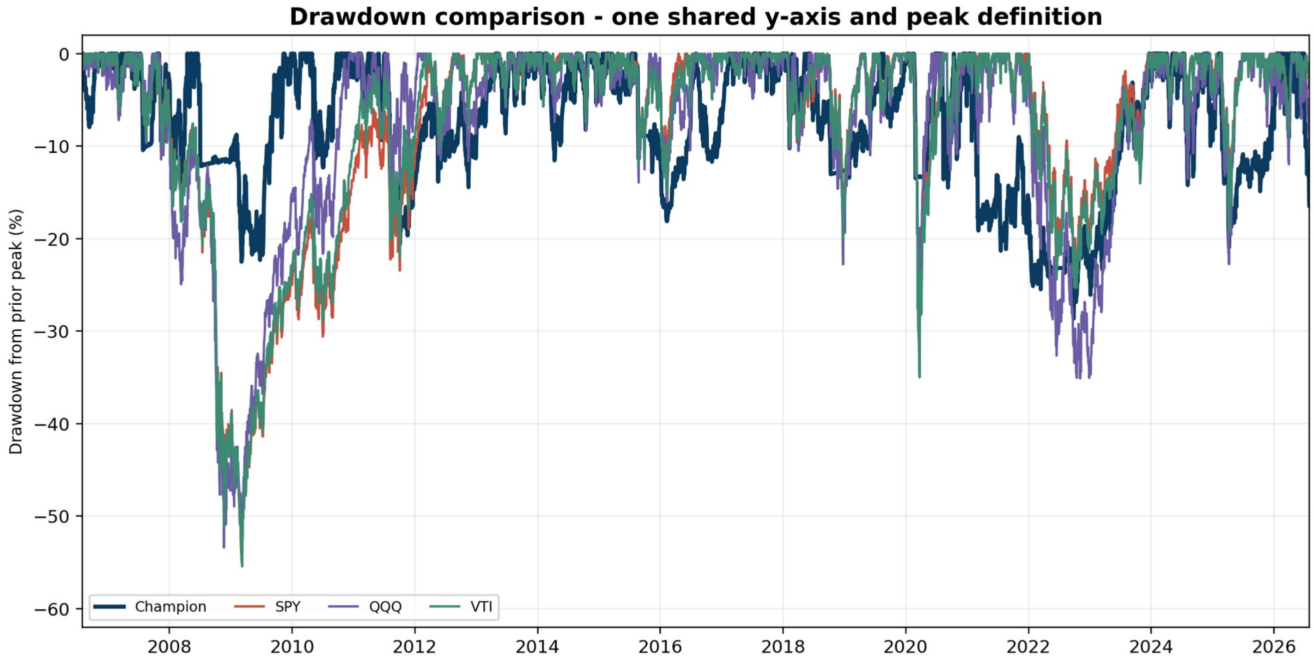

Figure 8. Drawdowns use the same zero line, shared y-axis, dates, and prior-peak definition.

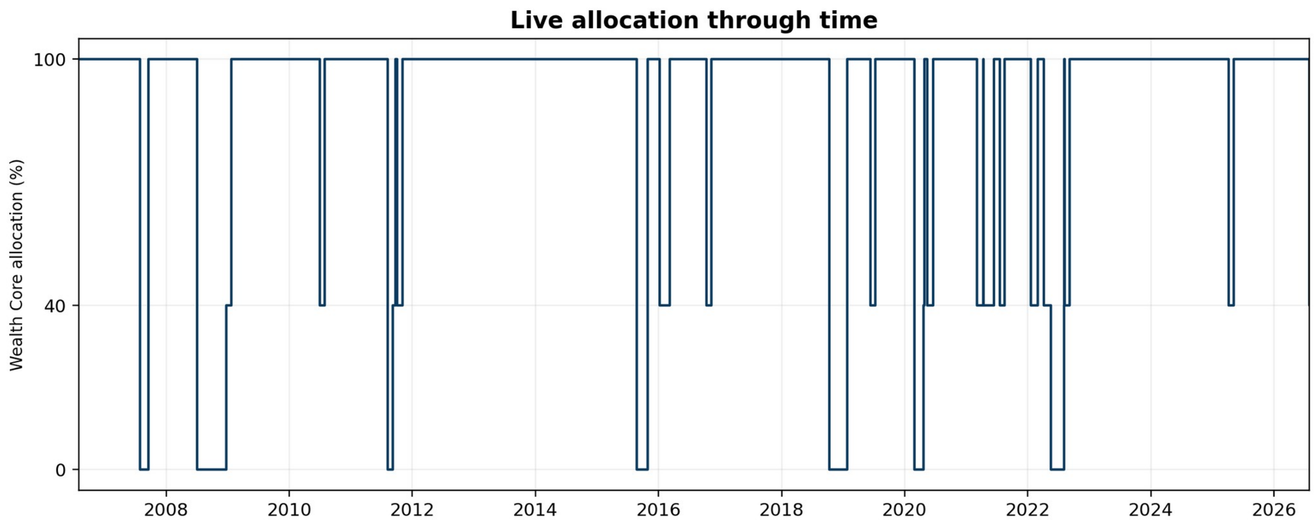

Figure 9. Wealth Core allocation through time. The controller uses only 100%, 40%, and 0% levels.

Risk is reduced, not eliminated

The strategy can lose money before protection activates, during overnight gaps, after recovery, or when a market decline does not satisfy the controller. The 15.5% ordinary-stress threshold intentionally permits material loss before de-risking because the system is designed to preserve long-term compounding rather than eliminate all volatility.

6. State machine and decision rules

6.1 Priority order

```
1. Maintain immutable shadow state and compute all features.
2. Evaluate SYSTEMIC_EMERGENCY.
3. If not systemic, evaluate PERSISTENT_BEAR.
4. Otherwise apply ordinary 15.5% backstop.
5. Evaluate recovery only for the currently active defensive state.
6. Persist decision and submit target allocation for next-session open.
```

6.2 Persistent Bear entry

```
ordinary_stress_active == true
stress_age_sessions >= 30
shadow_return_since_stress_start <= -0.02
shadow_return_40_sessions <= -0.03
damaged_breadth >= 0.75
green_breadth <= 0.25
```

6.3 Persistent Bear recovery

```
persistent_bear_age_sessions >= 20
healthy_confirmation_streak >= 5
shadow_return_20_sessions > 0.00
damaged_breadth <= 0.60
green_breadth >= 0.20
```

6.4 Ordinary stress

The ordinary controller becomes defensive when immutable shadow drawdown reaches or exceeds 15.5%. It targets 40% Wealth Core and 60% Treasury bills. Its recovery semantics must remain identical to the certified original controller unless explicitly versioned.

6.5 Systemic emergency

The systemic detector is the frozen confirmed-shock rule from the research champion. When active, it overrides the 40% ordinary floor and targets 0% Wealth Core. Its precise feature thresholds must be imported from the certified systemic configuration rather than inferred from this prospectus.

7. Immutable shadow portfolio

Why the shadow is essential

Once the live portfolio moves to 40% or 0%, its drawdown and recovery path no longer describe Wealth Core. The shadow preserves the counterfactual: what the strategy would have earned had it remained fully invested.

Required shadow state

Category	Required fields
Portfolio	cash, positions, shares, marks, NAV, high-water mark
Position episode	security ID, ticker, entry session, entry price, adjusted peak, review status
Control	slot cooldowns, security cooldowns, pending orders, corporate-action state
Derived	drawdown, 20- and 40-session returns, damaged breadth, green breadth
Audit	input hashes, state version, decision timestamp, prior and new state

Prohibited shortcuts

- Do not rebuild the shadow by buying current ranked names after a defensive episode.
- Do not reset entry dates or trailing peaks when live exposure returns.
- Do not use live NAV for shadow drawdown.
- Do not stop applying dividends, splits, mergers, delistings, or exits to the shadow.

8. Next-open execution specification

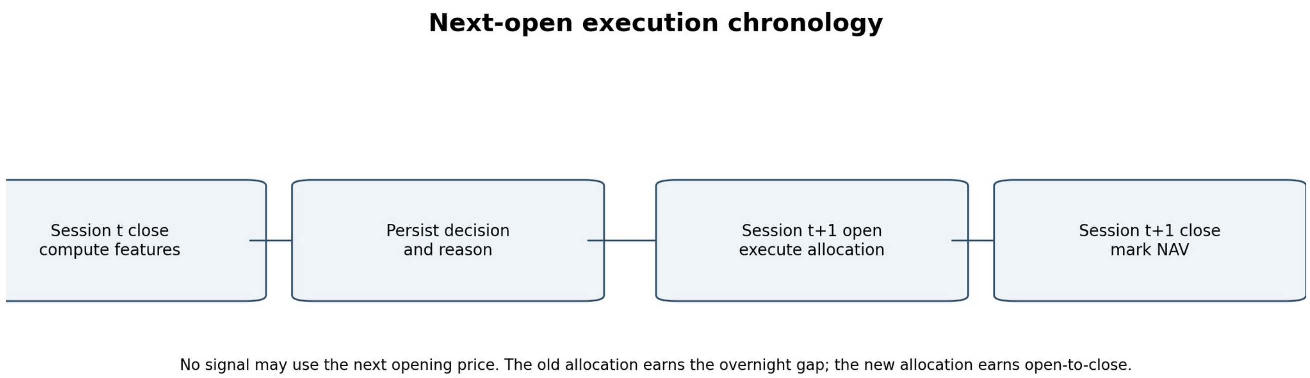

Figure 10. Decision and execution timeline. Signal information stops at the preceding close.

Return decomposition

For a transition decided after close t and executed at open t+1:

```
overnight_return = old_allocation * WC(close_t -> open_t+1)
                 + (1-old_allocation) * TBILL(close_t -> open_t+1)

intraday_return = new_allocation * WC(open_t+1 -> close_t+1)
                + (1-new_allocation) * TBILL(open_t+1 -> close_t+1)

cost = one_way_cost * abs(new_allocation - old_allocation)
portfolio_return = (1+overnight_return)*(1+intraday_return)*(1-cost)-1
```

Fill requirements

- Use actual adjusted opens for research replay and raw/as-traded opens for order accounting.
- Price all existing Wealth Core positions at the transition open before changing allocation.
- Defensive sleeve buys and sells occur at the same next-open event.
- If opening prices are missing, apply an explicit fill policy and log coverage; never substitute a later close silently.
- The validated reconstruction achieved 99.69% position pricing coverage across transition dates.

9. Accounting, costs, and defensive sleeve

Defensive asset

The research sleeve is a short-duration US Treasury-bill total-return proxy. Production may use BIL, SGOV, or a direct T-bill implementation only after instrument-specific execution, distributions, taxes, settlement, and liquidity are specified. The research result must not be assumed transferable without that mapping.

Costs

Cost item	Research default	Production requirement
Allocation transition	10 bps one-way on changed notional	Parameterize by account size and liquidity
Wealth Core internal trading	As represented in certified base path	Use trusted simulator fills and costs
Defensive distributions	Included in adjusted-price series	Post cash or reinvest consistently
Taxes	Excluded	Model separately by account type

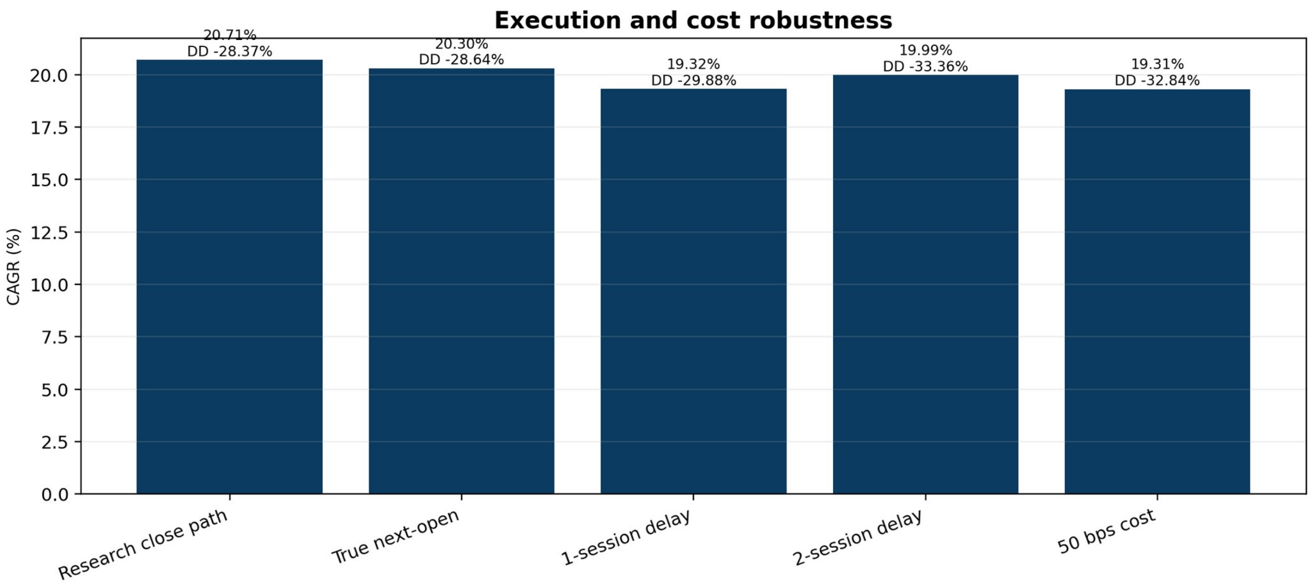

Figure 11. Robustness to execution delay and higher transition costs.

10. Data contracts and persistence

Input	Frequency	Minimum fields	Fail-safe
Trading calendar	Daily	session date, prior and next session	Do not invent sessions
Equity bars	Daily	security ID, raw O/H/L/C, adjusted close factors	Freeze transition or follow explicit missing-price policy
Defensive bars	Daily	raw and adjusted O/H/L/C	Retain prior allocation if unavailable
Corporate actions	Event	permanent ID, action type, ratio/cash, effective date	Apply to live and shadow identically
Wealth Core state	Daily	holdings, cash, peaks, reviews, cooldowns, pending	Abort on incompatible version
Controller state	Daily	state, age, stress anchor, streaks, reason	Atomic persistence before order submission

Controller snapshot schema

```
controller_state_version: string
as_of_session: date
current_state: NORMAL | ORDINARY_STRESS | SYSTEMIC_EMERGENCY | PERSISTENT_BEAR
current_state_age: integer
stress_start_session: date | null
stress_start_shadow_nav: decimal | null
healthy_confirmation_streak: integer
last_target_allocation: decimal
pending_target_allocation: decimal | null
feature_snapshot: object
decision_reason_codes: array[string]
input_lineage_hash: string
```

11. Reference pseudocode

```
def on_session_close(session, shadow, controller, config):
    shadow = wealth_core_step(shadow, session)
    f = compute_features(shadow, controller, session)

    if systemic_entry(f, config):
        desired_state = SYSTEMIC_EMERGENCY
    elif controller.state == PERSISTENT_BEAR:
        desired_state = recover_or_hold_persistent(controller, f, config)
    elif persistent_bear_entry(controller, f, config):
        desired_state = PERSISTENT_BEAR
    else:
        desired_state = ordinary_controller_state(controller, f, config)

    target = allocation_for(desired_state)
    decision = Decision(session, desired_state, target, f, reason_codes(...))
    atomic_persist(shadow, controller.updated(decision), decision)
    enqueue_next_open_target(decision)
    return decision

def on_next_session_open(open_event, persisted_decision):
    assert open_event.session == next_session(persisted_decision.session)
    mark_old_book_at_open(open_event)
    execute_changed_notional(persisted_decision.target_allocation, open_event)
    charge_transition_cost()
    persist_fills_and_live_state_atomically()
```

Deterministic comparison order

- Use decimal or consistently rounded floating-point calculations.
- Define whether thresholds are inclusive exactly as shown.
- Calculate session ages from eligible trading sessions, not calendar days.
- Compute returns from point-in-time NAV observations without forward filling missing shadow sessions.
- Sort holdings by permanent security ID before breadth aggregation to guarantee reproducibility.

12. Worked examples

Example A - 2022 Persistent Bear entry

Feature at decision close	Observed/required
Ordinary stress age	30 sessions or more
Shadow loss since stress start	At or below -2%
40-session shadow return	At or below -3%
Damaged breadth	At least 75%
Green breadth	At most 25%
Decision	Enter PERSISTENT_BEAR
Execution	Next session open; 0% Wealth Core

In the validated path, the signal was confirmed at the May 17, 2022 close and the full defensive allocation became effective at the May 18 open. The live portfolio absorbed the overnight gap because that gap was unknowable at decision time.

Example B - recovery

After at least 20 Persistent Bear sessions, each close is evaluated for shadow recovery. Five consecutive closes must satisfy the recovery-health definition. The next-open target then returns to the allocation required by the underlying ordinary controller, which may be 40% or 100%; the system must not automatically jump to 100%.

Example C - restart during an episode

A service restart on the 37th Persistent Bear session loads the persisted state age, stress anchor, shadow history, and confirmation streak. It must not restart the 20-session hold, reset breadth history, or regenerate an already submitted order.

13. Validation and robustness

Validation	Result	Interpretation
True next-open overlay	20.30% CAGR; -28.64% DD	Pass; modest degradation from close-path research
Leave-one-crisis-out	Same held-out paths across neighboring thresholds	Supports structural stability
Decade walk-forward	First block stable; second optimized choice unstable	Freeze economically justified parameters; do not auto-retune
One-session delay	19.32% CAGR; -29.88% DD	Operational timing matters
Two-session delay	19.99% CAGR; -33.36% DD	Risk deteriorates with delay
25 bps cost	20.18% CAGR	Moderate-cost pass
50 bps cost	19.31% CAGR	Conservative stress remains attractive
Simpler rule	Positive-day metric removed with no path change	Prefer simpler production rule

Promotion status

The strategy is a provisional implementation champion. The remaining gate is an integrated trusted-simulator replay in which Wealth Core orders, controller transitions, defensive fills, persisted state, cash, and ledger reconciliation are produced by one engine.

14. Operational controls and observability

Required daily telemetry

- Current state, state age, live allocation, shadow NAV and drawdown.
- Stress start date and return since stress start.
- 20- and 40-session shadow returns.
- Damaged and green breadth with numerator and denominator.
- All entry/recovery booleans and failed predicates.
- Pending transition, expected next-open session, actual fills, cost, and price coverage.
- Live-versus-shadow position reconciliation and state hash.

Required alerts

Severity	Condition
Critical	Shadow update failed, state snapshot incompatible, duplicate transition, cash reconciliation failure
High	Missing opening prices above tolerance, live-shadow holdings divergence, defensive asset unavailable
Medium	Delayed order, abnormal slippage, stale market calendar, breadth denominator change
Info	State transition, recovery streak progress, parameter version change

Idempotency

Decision creation and order submission require stable unique keys, for example portfolio_id + signal_session + target_state. Reprocessing the same close or open must not create duplicate orders or reset state ages.

15. Risks and limitations

Risk	Description
Model risk	Historical thresholds may not generalize to future market structure.
Small event sample	Persistent Bear activated only a few times; statistical independence is limited.
Gap risk	Protection cannot avoid losses between decision close and next open.
Whipsaw	Recovery may occur before a renewed decline.
Benchmark mismatch	QQQ, SPY, and VTI are passive funds with different objectives and exposures.
Liquidity	Selling many positions at one open may create slippage beyond research assumptions.
Tax	Frequent large reallocations can create taxable gains.
Data revisions	Vendor corrections can change historical features and transitions.
Concentration	Wealth Core returns rely materially on a relatively small set of exceptional winners.
Implementation divergence	A seemingly small difference in state age, adjusted price, or breadth definition can change decisions.

No guarantee

The controller is a drawdown-management system, not a guarantee of positive calendar-year returns or a maximum-loss limit. A portfolio stop is triggered after losses have occurred and may re-enter before a durable recovery.

16. Engineering acceptance tests

Unit tests

- Every threshold boundary at one tick below, equal, and one tick above.
- Session-age handling across holidays and missing sessions.
- Breadth handling for zero, one, and changing position counts.
- State priority when systemic and persistent conditions are simultaneously true.
- Minimum-hold and recovery-streak resets.
- Next-open decomposition and cost on partial allocation changes.

Golden-path scenarios

Scenario	Expected behavior
Normal correction below 15.5%	Remain NORMAL
Ordinary threshold crossing	40% effective next open
Fast systemic confirmation	0% effective next open; systemic reason code
30-session persistent failure	0% effective next open; persistent reason code
One-day recovery spike	Remain 0%
Five-session confirmed recovery	Return next open to underlying controller allocation
Restart before open	Execute exactly one persisted transition
Missing defensive open	Apply documented fail-safe; never fabricate price

Certification gates

- Full replay parity between batch, rehearsal, shadow, and live controller.
- Ledger cash identity holds every session.
- Transition state survives process and host restart.
- Corporate-action parity between live and shadow.
- Opening-price coverage and missing-data policy reported.
- Configuration and source hashes embedded in result package.

17. Frozen configuration appendix


```yaml
strategy_id: wealth_core_systemic_zero_persistent_bear_v1
base_strategy:
  name: wealth_core_review119_stop30
  individual_stop: 0.30
controller:
  ordinary_stress_drawdown: 0.155
  allocations:
    NORMAL: 1.00
    ORDINARY_STRESS: 0.40
    SYSTEMIC_EMERGENCY: 0.00
    PERSISTENT_BEAR: 0.00
  persistent_bear_entry:
    minimum_stress_sessions: 30
    maximum_return_since_stress_start: -0.02
    maximum_shadow_return_40: -0.03
    minimum_damaged_breadth: 0.75
    maximum_green_breadth: 0.25
  persistent_bear_recovery:
    minimum_state_sessions: 20
    confirmation_sessions: 5
    minimum_shadow_return_20: 0.00  # strict greater-than
    maximum_damaged_breadth: 0.60
    minimum_green_breadth: 0.20
execution:
  decision_time: official_close
  effective_time: next_session_open
  one_way_transition_cost_bps: 10
defensive_asset:
  research_proxy: BIL_total_return
state:
  persistence: atomic
  shadow_mode: immutable_full_exposure
```


Versioning rules

Any change to a threshold, inequality, feature definition, defensive asset, execution timestamp, cost model, or recovery rule creates a new immutable strategy version. Historical runs must retain the exact configuration and code hash used.

18. Glossary and disclosures

Term	Definition
Damaged holding	A shadow holding meeting the frozen impairment criterion used by the certified controller.
Green holding	A shadow holding meeting the frozen healthy/positive criterion.
Immutable shadow	Counterfactual portfolio that remains fully invested in Wealth Core regardless of live allocation.
Stress anchor	Shadow NAV and session recorded when ordinary stress begins.
Systemic emergency	Fast, broadly confirmed shock state defined by the certified systemic detector.
Persistent Bear	Prolonged state of negative shadow expectancy after ordinary protection has persisted.
Next-open	First eligible market open after a close-generated decision.
Changed notional	Absolute change in Wealth Core allocation multiplied by live NAV.

Disclosures

Backtested results are hypothetical. They do not reflect an actual account, may benefit from hindsight in research design, and exclude investor-specific taxes, fees, custody constraints, and operational failures. Benchmark comparisons are for context and do not imply identical objectives, risk, holdings, or investability. QQQ and VTI are included as widely used passive fund comparisons, not as claims of superiority to all investment alternatives.

The 20-year champion result is an allocation-overlay reconstruction using a certified Wealth Core shadow path and actual Sharadar opening prices on transition dates. It is not yet a single integrated production-engine replay. The strategy remains subject to implementation and model risk.

Engineering handoff

An engineer should be able to implement the controller from Sections 6-17 without using informal conversation history. Any ambiguity discovered during implementation must be resolved by creating an explicit versioned requirement, not by guessing.